

# CHAYMA BETTAHER

Final-Year Software Engineering Student | AI, Cloud & Distributed Systems

✉ bettaherchayma77@gmail.com ☎ +216 55 684 933 📍 Tunisia 🌐 chayma bettaher 🐙 github.com/Bchayma

🔗 Chayma Bettaher 🇫🇷 French (C1) -English (B2) -Arabic (Native)

📄 Open to relocation for 6-month international final-year internship opportunities.

## PROFILE

Final-Year Software Engineering Student with hands-on experience developing real-world software applications and working across full-stack development, DevOps, AI/ML, and AWS cloud technologies. Strong in algorithms, competitive programming, and analytical problem-solving, combining a solid mathematical foundation with practical engineering skills. Highly interested in research and innovative engineering solutions.

## SKILLS

**Programming** — Python, JavaScript, TypeScript, Java, PHP, C/C++

**Frontend & Mobile** — React.js, Next.js, Angular, HTML, CSS, Flutter, WordPress

**Backend & APIs** — Node.js, NestJS, Spring Boot, REST APIs, GraphQL

**Cloud & DevOps** — AWS, Docker, Kubernetes, Linux, Jenkins, GitHub Actions, Terraform, Istio, Nexus, CI/CD

**Databases** — PostgreSQL, MySQL, SQL Server, Oracle, MongoDB

**AI/ML** — CNNs, Autoencoders, KNN, SVM, Random Forest, Computer Vision, NLP, PyTorch, TensorFlow/Keras, Scikit-learn

**Data Engineering & BI** — Hadoop, ETL, SSIS, Data Warehousing, Power BI, Data Visualization

**Software Engineering** — Software Architecture, Data Structures & Algorithms, OOP, Git, GitHub, Agile, Scrum

## CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02)
- Fundamentals of Deep Learning
- AWS Certified Solutions Architect – Associate (in progress)

## PROFESSIONAL EXPERIENCE

**Full-Stack Developer Intern, NADA Groups**

07/2026

- Developed a full-stack healthcare management platform using Next.js, NestJS, and MongoDB.
- Designed and implemented REST APIs, JWT authentication, and role-based access control.

**IT Consultant, Optima Junior Entreprise**

2025 – 2026

- Development of websites using WordPress and WooCommerce
- Prospecting, negotiation, and client relationship management

**Cybersecurity & Machine Learning Internship, VOCALCOM**

08/2025

- Built a network anomaly detection system using Isolation Forest on 1,500+ Wireshark-captured packets.
- Engineered a Python data pipeline with pandas/NumPy for data cleaning, deduplication, feature extraction, and SQLite storage.
- Performed EDA and developed a Streamlit dashboard for interactive anomaly visualization and filtering.

**DevOps & Kubernetes Internship, LOOYAS**

06/2025 – 07/2025

- Built a 7-stage Jenkins CI/CD pipeline with GitHub and a Dockerized SSH agent.
- Containerized 6 microservices with Docker and managed images through Nexus Repository.
- Deployed a 9-service Kubernetes architecture and configured Istio Gateway, VirtualService, sidecar injection, and CORS.

## TECHNICAL PROJECTS

**Cloud Infrastructure Project — AWS**

- Architected a highly available AWS VPC across 2 AZs and 8 subnets with public, web, app, and DB tiers.
- Configured RDS MySQL, ALB, Auto Scaling, and 5 security groups for frontend/backend infrastructure.
- Integrated S3, CloudFront, CloudWatch, and CloudTrail for delivery, monitoring, and auditing.

**Deep Learning Project - Anomaly Detection with Convolutional Autoencoder**

- Built an image classification pipeline on 2,000 PlantVillage images across 4 classes, comparing machine learning and deep learning approaches.
- Extracted handcrafted image features (color, texture, shape) and achieved 98.25% accuracy with a Random Forest model.
- Trained and benchmarked 5 models (CNN, MobileNetV2, EfficientNetB0, and classical ML), reaching up to 98% accuracy.

**Intelligent Exam Management System**

- Developed a university exam management web application using Angular, Spring Boot, and PostgreSQL, with layered architecture, JPA/Hibernate, and REST APIs.

## EDUCATION

**Engineering Degree in Software Engineering, Faculty of Sciences of Tunis**

2022 – Present

